Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3471-DE01

Title: (BEng/MEng)

Materials 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

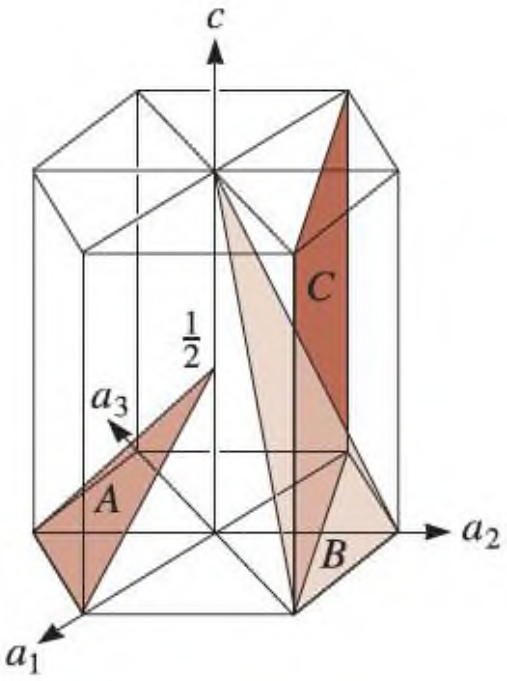
Revision:	

Question 1

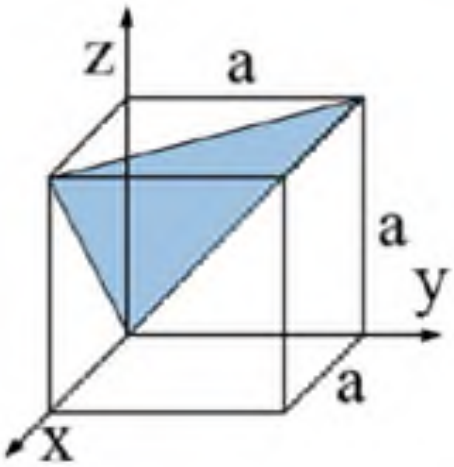
a) Determine the Miller indices of the planes in Figure 1a.

[30%]

I.



II.



III.

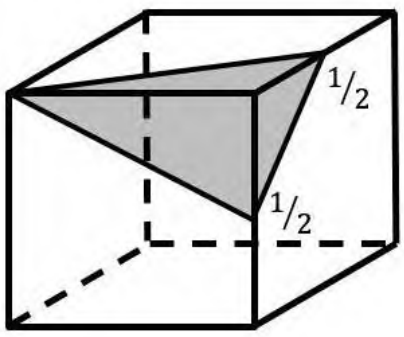


Figure 1a: Crystal Structures related to Q1a.

continued

b) Draw a Burgers circuit around the dislocation shown in Figure 1.b. Clearly indicate the Burgers vector that you find. What type of dislocation is this? In what direction will the dislocation move due to the applied shear stress? Reference your answers to the coordinate axes shown.

[30%]

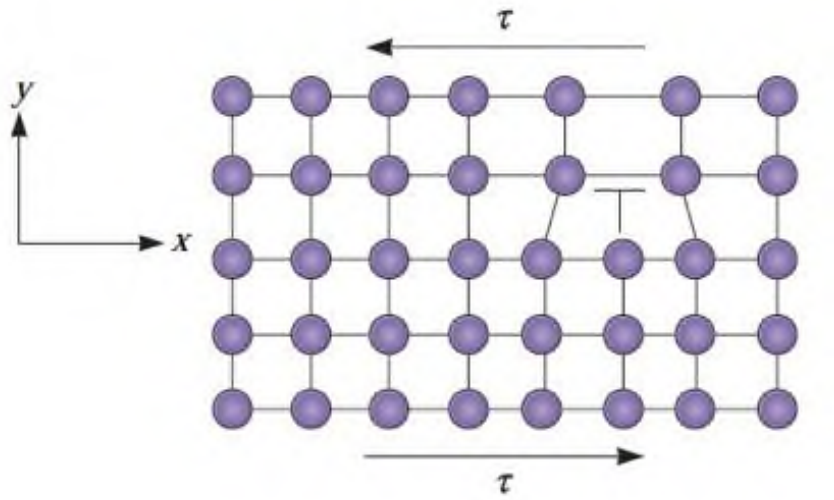


Figure 1b: Illustration related to Q1b.

c) A single crystal of a super alloy is oriented so that the [001] direction is parallel to an applied stress of 500MPa. Calculate the resolved shear stress acting on the (111) slip plane in the $[\bar{1}10]$, $[0\bar{1}1]$, $[10\bar{1}]$ and slip directions. Which slip system(s) will become active first?

[40%]

Question 2

a) When a Cu-Zn alloy solidifies, one portion of the structure contains 25 atomic percent zinc and another portion 0.025 mm away contains 20 atomic percent zinc. The lattice parameter for the FCC alloy is about 3.63 cm. Determine the concentration gradient in (a) atomic percent Zn per cm; (b) weight percent Zn per cm. Atomic weight of Zinc and Copper are 65.38g/mol and 63.55g/mol, respectively.

[10%]

b) Determine the number of three phase reactions in the graph below. Find the composition and ratio of the phases for every 3-phase reaction 10 degrees below the reaction.

[50%]

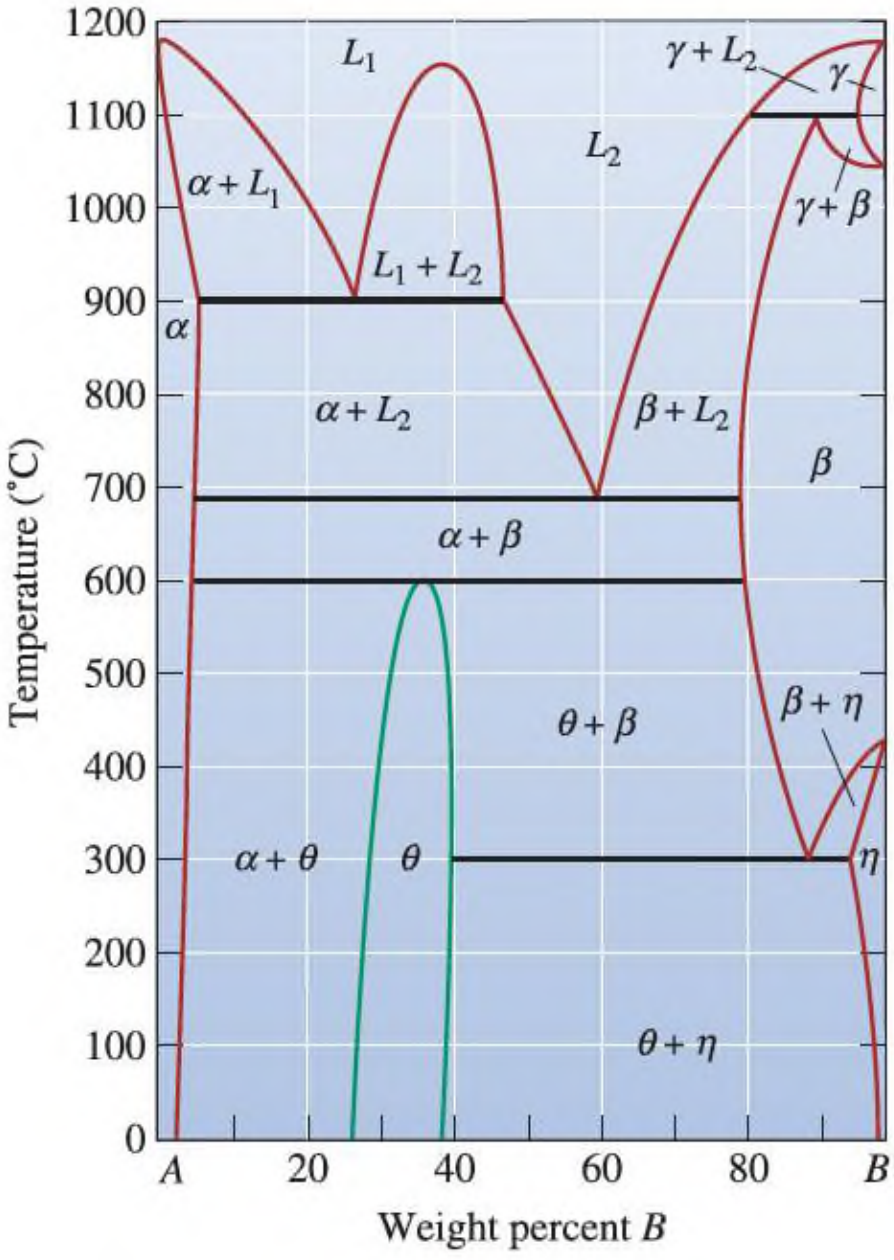


Figure 2b: Phase diagram related to Q2b.

continued

c) An engineer has found the following information about an unknown material (cf. Figure 2c and Table2c). He knows that is a simple cubic structure and its atomic radius is 2.34nm. Based on the following information and table below, identify the material.

[40%]

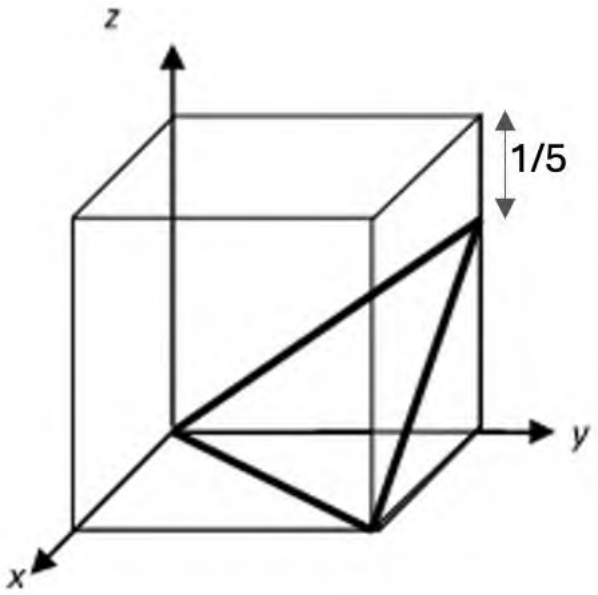


Figure 2c: Crystal structure and plane related to Q2c.

Name	**Interplanar Spacing (Å)**
AB3	**5**
C2Mo	**6**
DF23E	**5.5**
Xh43	**4**
Bh2I	**7.2**

Table 2c: Name and Interplanar spacing related to Q2c.

Question 3

a) At Durham, students can either attend lectures in person or watch lecture recordings (Encore). Discuss the environmental impact of both options and determine which is more environmentally friendly. Provide three reasons to justify your response.

[33%]

b) State the three main steps associated with material selection in the Design Process, with a brief description of each.

[9%]

c) An application requires a beam of 2-meter in length and for aesthetic reasons should be made of either Aluminium 6061 or soft wood. The two materials have strength constraint of 105 MPa. Using the quadrant graphs in Figure 3c. identify

i. The section area and density required for aluminium
ii. The section area and density required for soft wood
iii. The material with the lowest weight of the required beam

[24%]

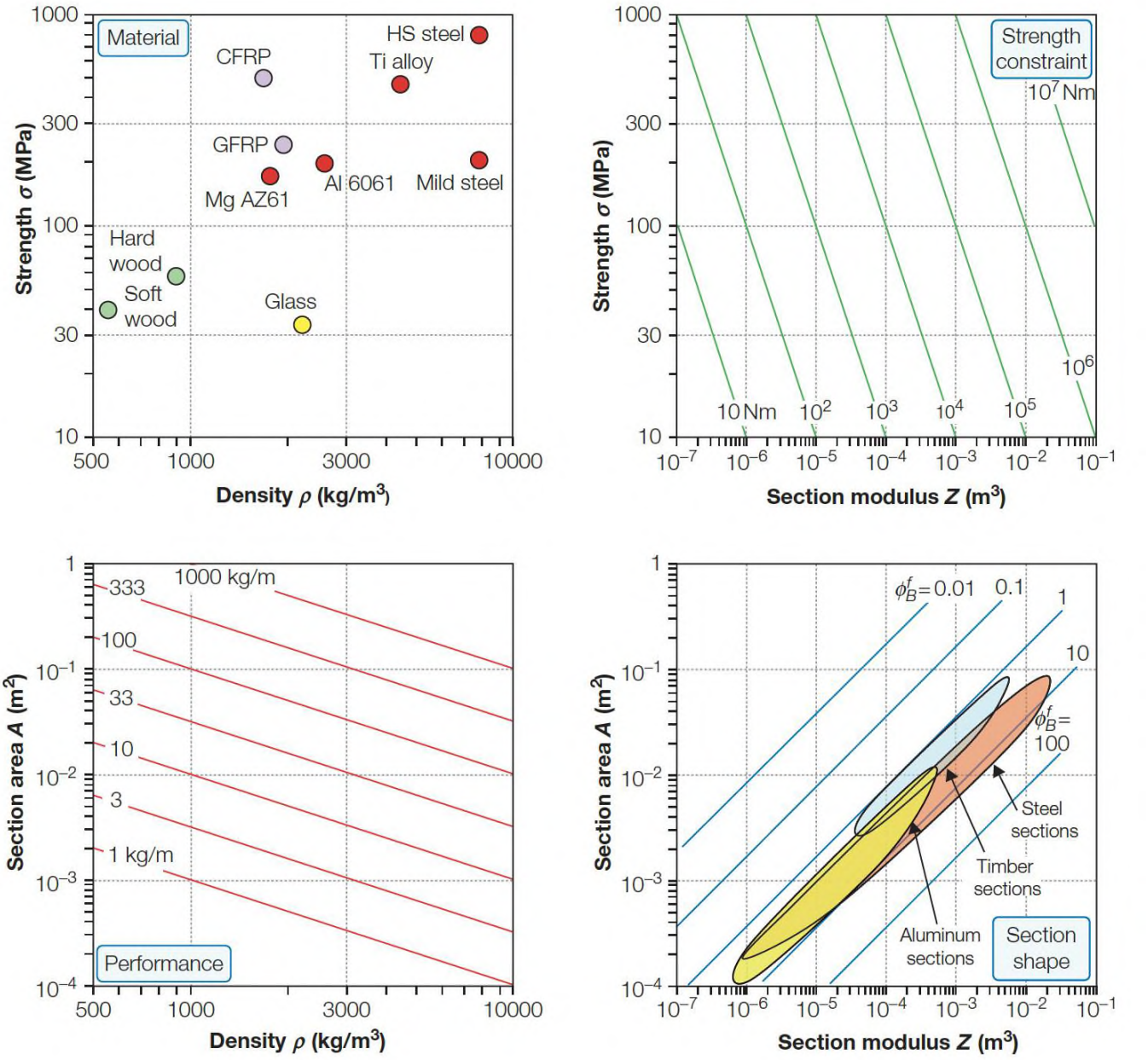


Figure 3c: Graphs related to Question 3c.

continued

d) Sunny Motor Manufacturing UK have forecasted the sales of their Juga car in 2025 will be 100,000 units. Gary Richardson is a body shop manager at Sunny's Coventry plant who oversees the welding operations to create complete car body shells in the body shop. He has prioritised the welding operations of car bonnets made with high strength steel of thickness 5mm in January 2025 for 1000 units. In your opinion, what welding type and welding process should Gary's team use to weld the car bonnets? Explain your reasoning in your own words with the aid of diagrams.

[34%]

Formula sheet

Bragg's Law:

$$sin\,\theta = \frac{\lambda}{2d_{hkl}}$$

Interplanar spacing:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Concentration of vacancies:

$$n_v = n\exp(-Q_v/RT)$$

ASTM grain size number:

$$N = 2^{n-1}$$

Percent cold work:

$$Percent\ of\ cold\ work = \left[\frac{A_0 - A_f}{A_0}\right] \times 100$$

Hall-Petch Equation:

$$\sigma_y = \sigma_0 + kd^{-1/2}$$

Diffusion coefficient dependency on Temperature

$$D = D_0\exp(\frac{-Q}{RT})$$

Stress relaxation:

$$\sigma = \sigma_0\exp(-t/\lambda)$$

Relaxation time:

$$\lambda = \lambda_0\exp(Q/RT)$$

The weight average molecular weight:

$$\overline{M_w} = \sum f_i M_i$$

The number average molecular weight:

$$\overline{M_n} = \sum x_i M_i$$

continued

Screen mesh for sorting particle sizes

$$PS = \frac{1}{MC} - t_w$$

Where PS = particle size, MC = mesh count, openings per linear mm and t_w = wire thickness of screen mesh.

Surface area (sphere)

$$A = \pi D^2$$

Where D = diameter of the spherical particle

Surface volume (sphere)

$$V = \frac{\pi D^3}{6}$$

Where D = diameter of the spherical particle

Area-to-volume ratio for a sphere

$$\frac{A}{V} = \frac{6}{D}$$

Where A = surface area, V = surface volume, D = diameter of the spherical particle

Area-to-volume ratio for any particle shape – spherical or non-spherical

$$\frac{A}{V} = \frac{K_s}{D} \text{ or } K_s = \frac{AD}{V}$$

Where K_s = shape factor; D = the diameter of a sphere of equivalent volume as the nonspherical particle.

continued

Power density

$$PD = \frac{P}{A}$$

where PD = power density, W/mm^2; P = power entering the surface, W; and A= surface area over which the energy is entering, mm^2.

Heat required to melt a given volume of metal

$$U_m = KT_m^2$$

where U_m = the unit energy for melting, J/mm^3; T_m = melting point of the metal on an absolute temperature scale, °K(°R); and K = constant whose value is 3.33 x 10^{-6} JK^{-2}mm^{-3}, when the Kelvin scale is used (and K = 1.467 x 10^{-5} JR^{-2}mm^{-3} for the Rankine temperature scale).

Heat energy available for welding:

$$H_w = f_1 f_2 H$$

where H_w = net heat available for welding, J, f_1 = heat transfer factor, f_2 = the melting factor, and H = the total heat generated by the welding process, J.

Energy balance equation

$$H_w = U_m V$$

Where H_w = net heat energy used by the welding operation, J; U_m = unit energy required to melt the metal, J/mm^3; and V = the volume of metal melted, mm^3.

Rate balance equation

$$R_{Hw} = U_m R_{WV}$$

Where R_{Hw} = rate of heat energy delivered to the operation for welding, J/s = W; and R_{WV} = volume rate of metal welded, mm^3/s.

$$R_{Hw} = f_1 f_2 R_H = U_m A_w v$$

here f_1 and f_2 are the heat transfer and melting factors; R_H = rate of input energy generated by the welding power source, W; A_w = weld cross-sectional area, mm^2; and v = the travel velocity of the welding operation, mm/s.

continued

Time to complete a single layer in rapid prototyping

$$T_i = \frac{A_i}{vD} + T_r$$

Where T_i= time to complete layer i; A_i = area of layer i; v = average scanning speed of the laser beam at the surface; D = diameter of the laser beam at the surface, and T_r = repositioning time between layers.

Build cycle time (stereolithography)

$$T_c = \sum_{i=1}^{nl} T_i$$

Where T_c = the STL build cycle time; and nl = the number of layers used to approximate the part.

END